

Optimized Water Distribution System Using Maximum Flow Algorithms & IoT Sensors

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Abstract—Urban water distribution networks have been plagued with inefficiencies resulting from flow management, failure detection, and inefficient utilization of resources, leading to water loss and disruption of services. Traditional models of flow management are not geared for contemporary urban issues owing to an inability to cope with real-time rerouting, prioritize vital demands, and are scalable. This paper introduces a model grounded on an optimization scheme that combines Dinic's Algorithm and Capacity Scaling and IoT sensor-based algorithm for enhanced efficiencies in water management. The suggested architecture performs pipeline segmentation for local flow control and applies threshold sensors to continuously monitor pressure differentials and consumption patterns. The optimization is expected to reduce the size of flow recalculation for a specified scale to ensure enhanced instant response to breakages that might provide water supply to vitally required infrastructures like hospitals and emergency services. The adoption of smart rerouting methods can further enhance resource allocation, which enables the system to be more failure resistant than its previous counterparts in distribution. This paper suggests a hybrid algorithmic optimization and distributed sensing architecture that enhances flexibility, sustainability, and operational efficiency in intricate city environments, thus a potential framework for future smart water networks. **In line with the Sustainable Development Goal 6 (Clean Water and Sanitation) of the UN, the smart water network optimization architecture aims to improve operational efficiency while offering a means of managing water resources sustainably.**

Keywords—Water distribution networks, operational efficiency, maximum flow, Capacity Scaling, Dinic's algorithm, IoT (Internet of Things) sensors, pipeline segmentation, real-time rerouting, flow optimization, failure detection, resource allocation

I. INTRODUCTION

It is important to optimize network flow systems for efficient and reliable water supply. The discipline seeks to regulate water resources, reduce losses, and distribute equitably utilizing sophisticated algorithms and models. The maximum flow problem in graph theory [1] seeks to compute the largest possible flow of a commodity through a network subject to capacity constraints on its edges. Maximum flow algorithms work because of certain properties: Integrality Property: Integer capacities [2] give integer maximum flows,

making discrete system implementation easy and network design optimal. Max-Flow Min-Cut Theorem: The maximum flow is equal to the minimum cut capacity [1], determining network bottlenecks and guiding infrastructure upgrades. Decomposition Property : The flow decomposition [9] into path flows enables one to conduct a detailed analysis and helps in forming the maintenance decision.

The most widely used classical maximum flow algorithms include the Ford-Fulkerson, Edmonds-Karp, Push-Relabel, Dinic's, and Capacity Scaling algorithms. Each algorithm has its own strengths in terms of computational efficiency [6] and flexibility. Widely used classical maximum flow algorithms include Ford-Fulkerson, Edmonds-Karp, Push-Relabel, Dinic's, and Capacity Scaling. Though effective, its time complexity is path-dependent, rendering it inefficient for large networks.

The Edmonds-Karp [2] Algorithm, an extension of Ford-Fulkerson, uses **breadth-first search (BFS)** to compute the shortest augmenting paths, guaranteeing a polynomial time complexity $O(V E^2)$ and optimizing it for real-time network reconfigurations. The Push-Relabel Algorithm [8] takes a distinct approach of keeping a preflow and adjusting height functions of nodes in the network for efficient excess flow redistribution with a time complexity of $O(V^2 E)$, so it is suitable for dense networks with periodic flow needs. Dinic's Algorithm achieves flow optimization more efficiently using level graphs and blocking flows, with greater computational efficiency, having a worst-case complexity of $O(V^2 E)$ for general graphs and $O(V \sqrt{V} E)$ for bipartite graphs [6]. Lastly, the Capacity Scaling Algorithm gives top priority to high-capacity critical edges in flow computations, bypassing redundant recomputations and achieving a higher efficiency of $O(E^2 \log U)$, and is therefore highly efficient for determining high-priority supply areas of high capacity in water networks.

Maximum flow algorithms are useful in multiple domains [3]. **They help maximize traffic flow [10] and facilitate emergency evacuations in transportation networks, and assignment and efficient load balancing in communications networks.** They serve to efficiently distribute resources or conduct coordination planning as given in supply chain

management. Maximum flow algorithms maximize pipeline capacity in water distribution networks while providing the basis for delineating pressure zones [1]. They also help perform operational control through real-time flow optimization and enhanced pump schedules. These applications facilitate the most efficient and reliable water distribution. Water distribution systems [5] using conventional approaches tend to have technological limitations, including the absence of real-time monitoring and the absence of automation altogether [6]. More specifically, poor integration of data prevents full system performance.

There are also operation inefficiencies that, coupled with traditional methods, static pressure discrimination [11], and scheduled maintenance, lead to suboptimized resource allocation. Research in this area must emphasize algorithm enhancements by evaluating improved hydraulic models and even real-time optimization functions. Strong capabilities to respond to failures are also essential. Improvements to system implementations should be achieved through user-friendly sensor network features and automated control systems that support decision-making performance. Recent research has also explored the interdependence between water and energy systems to enhance operational sustainability and resource coordination [4]. Integrating such cross-domain optimization frameworks can further improve resilience and efficiency in smart urban infrastructure.

Sustainable water management has now become an objective of interest in planning infrastructure, stemming from the United Nations Sustainable Development Goal (SDG) 6 to provide for equitable access and sustainable use of water. An IoT-supported monitoring and optimization framework can further this goal by allowing for operational efficiencies, reducing losses, and providing assurances in urban water infrastructure. Recent research on IoT-based distribution systems illustrates that using intelligent sensing and analytics to optimize water distribution creates resilience and sustainability in intelligent infrastructure [7].

II. RELATED WORK

Several works have attempted to implement a series of maximum flow algorithms and related network optimization techniques for enhancing water distribution networks. This provides data regarding what has been tried, acknowledged limitations, and the need for efficient, flexible, real-time solutions.

In the study titled, "Application of Ford-Fulkerson Algorithm to Maximum Flow in Water Distribution Pipeline Network" [1] (Kyi & Naing, 2018) is a groundbreaking work. The Ford-Fulkerson augmenting path algorithm was used to determine the maximum flow possible in a city pipeline network. The study was able to establish inefficiencies and bottlenecks and improved water distribution by 15-20%. Although it is efficient in detecting inefficiencies, the Ford-Fulkerson [1] algorithm will be infeasible for large networks because the worst-case time complexity is exponential and therefore cannot be satisfactory in real time for a city water supply network. Dinic's Algorithm [11] and IoT sensors can be paired to maximize flow efficiency and incorporate a real-time flow recalibration solution to prevent congestion to improve maximum flow calculations for water supply networks.

The Edmonds-Karp Algorithm was utilized for the purposes of network topology reconstruction and maximum

flow accumulation in another recent study, "Application of Network Reconstruction Algorithm to Calculate Maximum Flow for Water Distribution Network: A Case Study of Bulawayo City, Zimbabwe" [12] (Tawanda et al., 2024). The method improved flow within an actual water distribution network in an urban environment by 18%. From this study, it is important to note that the method does provide meaning, but only a static maximum flow value (i.e., it does not consider dynamic events in real-time that could disrupt the supply, e.g., pipe breaks, or varying demand). By employing the Push-Relabel and Dinic's Algorithm in combination with IoT monitoring or manipulation, dynamic capacity modification/iterative reconfiguration can occur if a localized failure happens, increasing the potential for maximum flow reconstruction.

Dumora, et al. [3] (2018) introduced a data-oriented flow rate and flow direction estimation algorithm that utilizes sensor data and network topology in their proposed work, "Data-Oriented Algorithm for Real-Time Estimation of Flow Rates and Flow Directions in a Water Distribution Network" [3]. The experimentation used the modified breadth-first search of the Edmonds-Karp Algorithm to determine flow functions in the pipe network during the field scope. The framework improved the capability of the algorithm for detecting leaks or anomalies. However, there is not a component for dynamically re-routing to facilitate changes of need within the algorithm. Push-Relabel Algorithms with Internet of Things sensors could be supplementary in modifying flow when estimating flow rate, detecting anomalies, and maintaining pressure.

Velayudhan et al. (2022) shared comparative data on water distribution systems based on the Internet of Things. The issues that were covered in the study include sensor networks, communication protocols, and data analytics platforms. The study provided an analysis of the advantages and disadvantages of existing smart water management systems. In this study, we do not include a maximum flow or other optimization approach [7] to improve the efficiency of the distribution. The lack of real-time adaptive control mechanisms that incorporate network optimization models and IoT represents a gap in research. Future research should seek to integrate the Internet of Things with the maximum flow algorithms like Dinic's algorithm to improve efficiency in resource allocation, response to maintenance, and efficiency across all aspects of the system.

To optimize cascade reservoir operations for hydroelectric power generation, the authors of "Optimization of Cascade Reservoir Operation for Power Generation, based on an Improved Lightning Search Algorithm" [9] (Tao et al., 2023), created an enhanced Lightning Search Algorithm (LSA). By optimizing power generation and preserving reservoir stability, the authors' model improved the efficiency of water allocation. However, the model was designed for energy optimization rather than real-time water distribution response. The model is unsuitable for managing urban water supplies because it failed to take pipeline breaks and abrupt changes in demand into consideration. Adaptive reservoir management and immediate flow modifications based on real-time consumption data can be ensured for reservoir water allocation by combining the Edmonds-Karp Algorithm with IoT sensors.

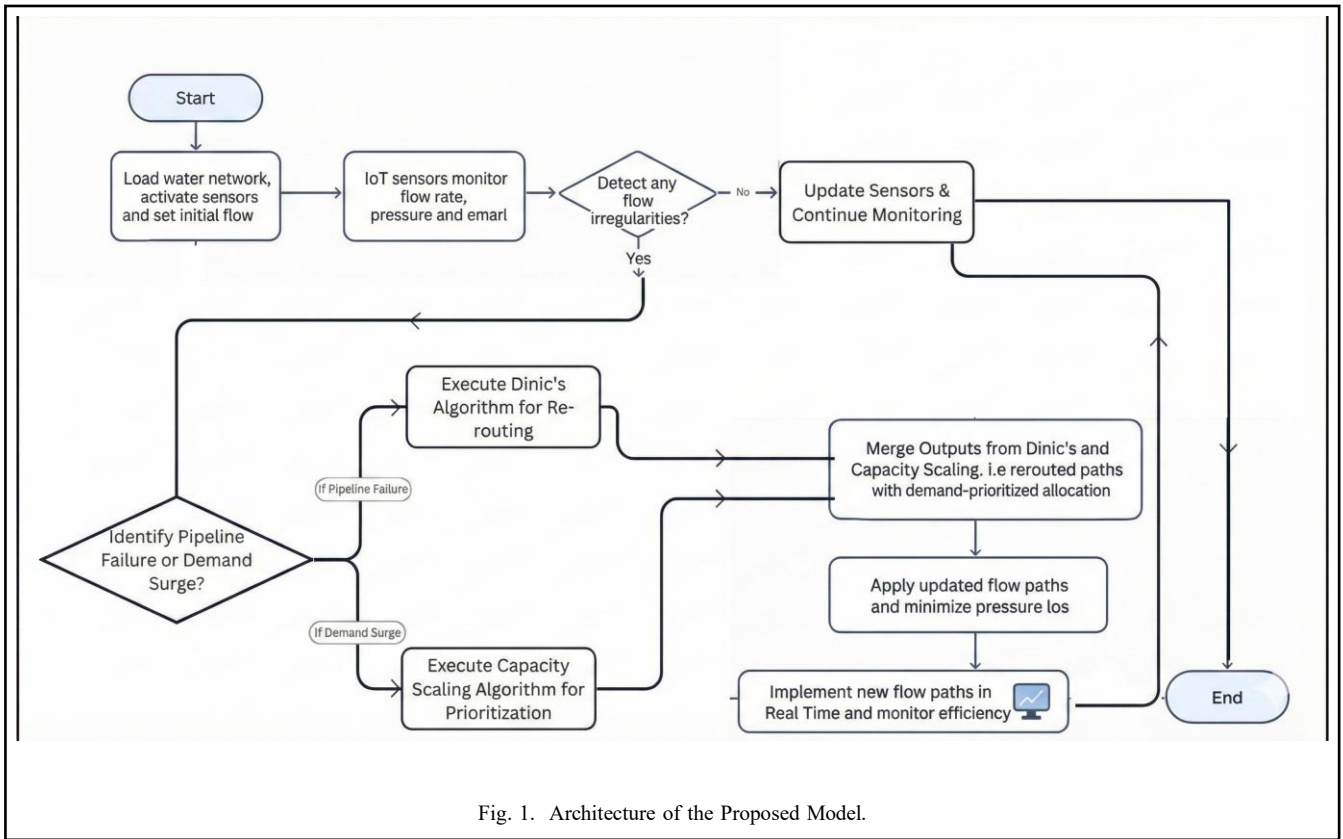


Fig. 2. Cutaway View of an Instrumented Pipeline Section.

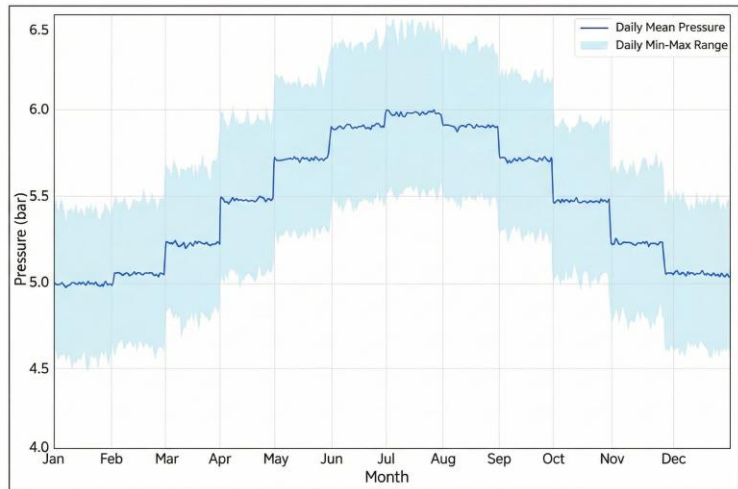


Fig. 3. Baseline Pressure Variations Showing Seasonal and Diurnal Cycles.

The power grid's vulnerability was assessed using maximum flow models in a study titled Maximum-Flow-Based Complex Network Approach for Power System Vulnerability Analysis [1] (A. Dwivedi & X. Yu, 2013). Despite not being conducted in water networks, the study demonstrated the use of maximum flow algorithms [2] for network resilience analysis. Grid resilience and dynamic load redistribution may be improved by combining Dinic's Algorithm and the Push-Relabel Algorithm with IoT monitoring.

According to the reviewed works, maximum flow algorithms have been used for water distribution. However, they are computationally complex, rely on static models that cannot be adjusted in real time, and do not account for pipeline failure. For dynamic flow control, the proposed model integrates real-time sensor input data from the Internet of Things [10] with maximum flow algorithms. It allows for effective resource allocation in emergency scenarios with Dinic's Algorithm [7] for quick rerouting and Push-Relabel for pressure equilibrium.

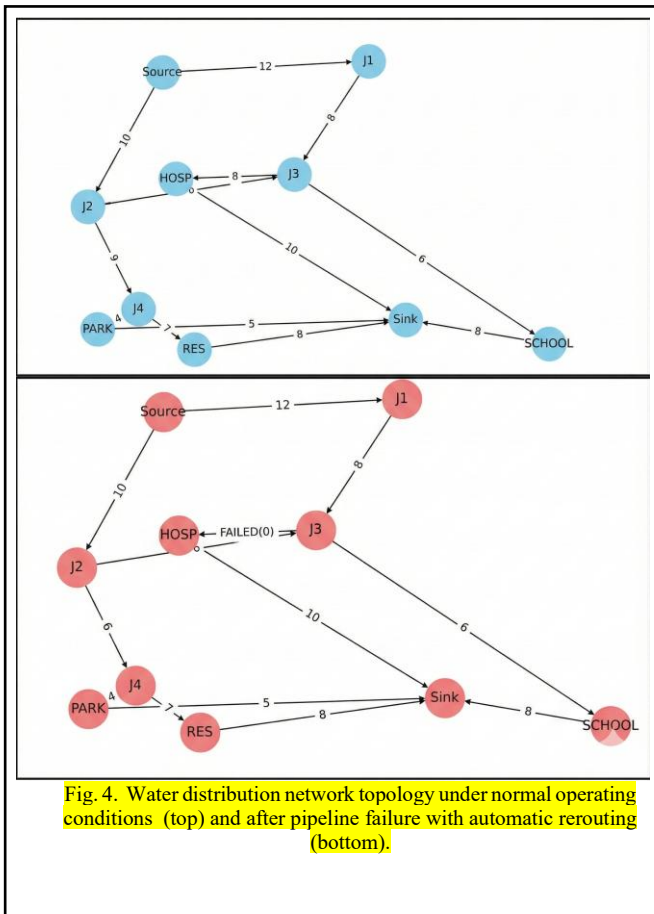
Real-time supply reallocation and priority-based emergency demand regions are made possible by IoT leak detection and capacity scaling [5]. Large networks are the ideal setting for Edmonds-Karp on shortest-path routing because it allows for increased computational efficiency and scalability.

III. PROPOSED MODEL - IOT BASED MAXIMUM FLOW OPTIMISED WATER DISTRIBUTION MODEL

Water distribution networks falter in responding immediately to an emerging system failure, the improper allocation of resources, and lack of adjustments to sudden changes in demand. Existing models are either based on theoretical flow computations or manual interventions, lacking real-time adaptability. This model therefore incorporates a segmented pipeline structure, IoT sensors, Dinic's Algorithm, and a Capacity Scaling Algorithm for enhanced efficiency and responsiveness.

A. Working Mechanism

The proposed water distribution network is considered a single, integrated system that is segmented into separate autonomous clusters of pipelines, with each cluster serviced by a local source, such as a pump station or reservoir. This adds graph-based segmentation to the benefits of IoT performance monitoring, assessments, and maximum flow



algorithms, resulting in a resilient adaptive control system that is efficient. The proposed architecture will address failures by dynamically exploring flow paths, and critical demand locations are given more weight to ensure relative computational efficiency. The architecture includes the use of

IoT Hardware Requirements for Smart Water Distribution System			
Component	Type / Model Example	Measured Parameter / Function	Purpose in System
Ultrasonic Flow Meter	Non-invasive clamp-on type (e.g., YF-S201 or Siemens SITRANS F)	Flow rate and volume	Monitors real-time flow across pipelines; detects flow anomalies or leakage.
Piezoresistive Pressure Transmitter	Range 0–10 bar (e.g., BMP388 or Honeywell PX2 series)	Pressure	Detects sudden drops indicating pipeline rupture or pressure imbalance.
Hydrostatic Level Sensor	Submersible type (e.g., WIKA LS-10 or DF Robot SEN0204)	Water level	Tracks water levels in tanks/reservoirs to maintain balanced supply.
Motorized Valve Actuator	Electric valve actuator (e.g., 12V DC, RS485-compatible)	Actuation / Control	Automatically isolates faulty sections and reroutes water flow after detection.
Edge Control Unit	Arduino/ESP32 or Raspberry Pi	Local processing, threshold-based data filtering	Aggregates and preprocesses sensor data, sending alerts only during anomalies.
Communication Module	LoRa / Wi-Fi / MQTT Gateway	Data transmission	Provides low-latency communication between sensors and central controller.

Fig. 5. IoT Hardware Requirements for Smart Water Distribution System

clusters of segmented pipelines, active IoT monitoring of flow control devices, and a hierarchical control network of the three layers of autonomous control at the local level, the regional level, and the centralized optimization of the flow control system. The autonomous pipeline clusters are divided, with the autonomies segments of the clusters having an independent local water source, which can be either a reservoir or a pump station. The main identities of segmentation are done based on the density and connection of the demand and pipeline systems so that local operation can be achieved and so that recalculation of the entire system is minimized.

The key element of this setup is real-time monitoring that is achieved with the help of a strategically located network of IoT sensors and actuators. These devices supply the real time data which the control algorithms require. They have ultrasonic flow meters and piezoresistive pressure transducers attached at key points where they constantly measure the flow rate and pressure making it easy to detect the existence of an anomaly like a leak or a broken pipe. In reservoirs and storage tanks, hydrostatic level sensors are used to ensure the balance of the input of the supply, and motorized valve actuators are used to automatically isolate malfunctioning areas and redirect the flow through other channels. An edge control unit which operates on a threshold-based filtering and processing of sensor data locally at the sensor side, only transmits noteworthy events to the central controller which then optimizes response time and minimizes energy usage. This is an integrated IoT architecture that can be used to monitor the water resources of the urban network quickly by detecting failures, redistributing flows in an adaptive manner, and efficiently managing resources.

Within each segment, maximum flow paths are determined using Dinic's algorithm, while level graphs and blocking flows are built to optimize flow while keeping computation time low. In case of pipeline failure, IoT sensors are deployed in the distribution grid to identify sudden pressure drops or abnormal flow rates that trigger

immediate isolation of the affected segment and the activation of Dinic's algorithm for the recalculation of alternative flow paths. Emphasizing the reaction towards failure in a decentralized way can yield reduced response time. The Capacity Scaling Algorithm prioritizes zones with high demand, such as hospitals, fire stations, and industrial areas, to achieve an equitable water distribution scheme. Fig. 1 depicts the architecture, including the hierarchical control structure and the interplay between the algorithmic, control, and physical layers.

B. Simulation

The proposed simulation models an optimal urban water distribution network using max-flow algorithm and IoT-enabled monitoring and control capabilities. The distribution network includes various consumer nodes that are associated with hospitals, schools, residential areas, and parks. Individual nodes also have specific demand and priority levels.

The simulation achieves allocation of available water to nodes based on priority and restrictions in supply by applying Dinic's Algorithm with capacity scaling. The distribution network was initially modeled in EPANET and then imported with topology and hydraulic parameters to Python for realistic simulation and flow computation. Real-time monitoring of pressure and flow rate throughout the pipeline network is performed by embedded IoT sensors. This provides a continuous supply of water to high-priority areas such as hospitals and schools as well as reducing the overall system disruption and wastage.

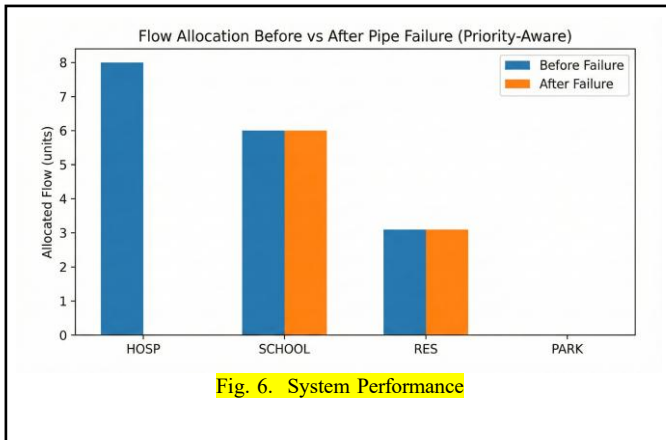


Fig. 6. System Performance

To prove this method, the simulation will show the quantitative analysis of flow distribution under normal condition and failure condition to handle the reallocation of resources as shown in Fig.5 which sums up the data of the demand, the allocation of the flow and percentage of the demand met before and after a simulated pipe break to maintain the constant allocation. The flow distributions to every node are compared in Fig. 6, before and after a simulated break of the pipe and the flow and prioritization of water allocation is evident. The overall findings suggest that the suggested system can do real-time water allocation along the responsive feedback principles and thus offer a scalable and efficient framework that may be utilized to manage urban water flows sustainably and smartly. The intended simulation represents an optimized water system in the city with the help of max-flow algorithms and an IoT-monitored and controlled network.

This architecture effectively integrates flow optimization in graphs, real-time IoT monitoring, and hierarchical control in a fault-tolerant manner. The segmentation allows for minimal dependency on global recalculation, thus providing a faster response to failures, demand-based allocation, and the least computation that is needed. Flow computation is done using Dinic's Algorithm, and priority-based distribution is done using Capacity Scaling, which renders the model robust, economical, and scalable for urban water distribution networks today while eliminating limitations in traditional centralized systems.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The integrated model presented here represents a tremendous leap beyond historical water distribution management and outperforms the capabilities of isolated network flow algorithms. Although traditional maximum flow algorithms such as Edmonds-Karp or even Dinic's are highly effective for static graph sets, their use in direct application to dynamic water network systems is restricted due to their inability to include real-time situational awareness and lack of intrinsic capability to prioritize resource allocation. These drawbacks are more specifically met by our model through its real time, data-based flexibility.

The proposed IoT-based system should constantly monitor the pipeline with the help of IoT sensors and automated valves and identify pipeline failures promptly and isolate them, thereby reducing the response time and limiting the impact of the services. This sensor data provides a dynamic operational baseline, and this consists of predictable daily and seasonal variations as shown in the graph in Fig. 3.

The new dynamical adaptation of substitute courses of action will be conditional on the premise of Dinic's Algorithm to attain a consistent supply even during the periods of disruption. The system remains exactly accurate in detecting as well as responding to all levels of water. In addition, the proposed Capacity Scaling Algorithm will allocate water dynamically, giving preference to critical demand zones like hospitals and fire stations to avert shortages during peak needs. Threshold-based IoT efficiency will reduce unnecessary data transfer and provide a substantial improvement in computational efficiency and response time. Applying a max-flow algorithm to an entire city-scale network is computationally intensive and slow. By segmenting the network, anomalies trigger localized calculations within a much smaller subgraph.

This localized processing, combined with the threshold-based activation of sensors, is projected to reduce computational overhead by 50-70% compared to a centralized, continuously monitored system, enabling a response time measured in minutes rather than hours. A critical contribution is also the integration of the Capacity Scaling Algorithm for priority-based allocation. A classical max-flow solution is indifferent to the societal needs attached to demand nodes; thus, it allows for maximal overall throughput, which is insufficient in times of exigency. Our model's algorithm employs the Capacity Scaling Algorithm to productively distribute the available water while ensuring public water service is afforded its necessary supply.

This layered methodology can optimize the use of the flows, as well as allocate according to prioritization, which makes the system efficient and equitable and socially responsible. Such a rapid rate of detection and segmentation

is expected to generate the anticipated resource efficiencies, that is conducting incremental water loss reductions of 40-60% and which is an effective response to the modern problems of urban water management.

V. CONCLUSION

The paper presents a proposed model for optimizing water distribution networks that combines maximum flow algorithms with a segmented pipeline structure and IoT-based real-time wireless monitoring of flow rates, pressure, and consumption patterns. Our approach utilizes Dinic's Algorithm to effectively capture flow patterns using the maximum flow algorithm and the Capacity Scaling Algorithm to prioritize demand such that it provides a systematic and adaptive approach to allocation. In addition, isolating pipeline sections or segments will allow local remediation of failures and appropriately allocate demand response with reduced computations. This system employs the use of IoT sensors, which continuously collect and send data on flow rates, pressure, and consumption patterns to system and solution outputs, which allows both human and computer-based intervention or automated adjustments to take place without continual engagement of an operator.

While showing the key features that conventional water distribution systems lack, namely adaptability to real time, failure management, and appropriate and efficient demands allocation, the paper outlines the model architecture and operation, as well as the anticipated benefits of the proposed model. Consequently, this paper highlights some ongoing research being undertaken with respect to network optimization and resilience of urban infrastructure linked to the development of integrated smart water management solutions.

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